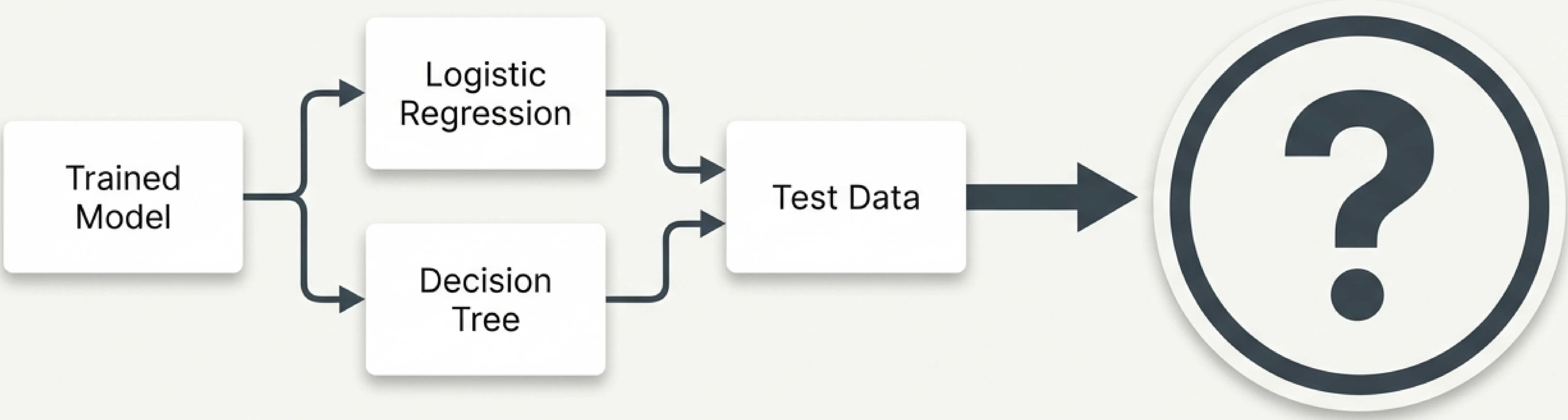
Beyond the Prediction

A Visual Guide to Evaluating Classification Metrics.

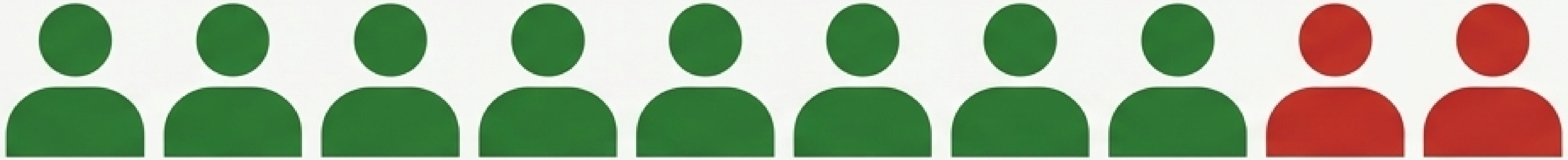
Your model is trained. But is it any good?

After running testing data through a classification model, we must objectively measure its performance to know if it is ready for the real world.



Level 1: Accuracy (The Intuitive Metric)

$$\textbf{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}}$$



8 correct predictions out of 10 = 80% Accuracy.

Context dictates what makes accuracy good

High Stakes

A 99% accuracy rate for predicting cancer is dangerously low if the 1% error rate costs lives.

Low Stakes

An 80% accuracy rate for predicting weekend takeaway orders is perfectly acceptable.

Accuracy hides the nature of the mistake

A single number tells us that mistakes happened, but not what kind of mistakes they were.

1. Predicting a healthy person is sick.

In a heart disease prediction model, a 10% error rate contains two very different scenarios.

2. Predicting a sick person is healthy.

Level 2: The Confusion Matrix

To understand the specific nature of our model's mistakes, we map what the model predicted against what actually happened.

		Predicted Values	
		0	1
Actual Values	0		
	1		

Decoding the Terminology (The Cheat Code)

True

Positive

This tells you if the model was actually right in reality.

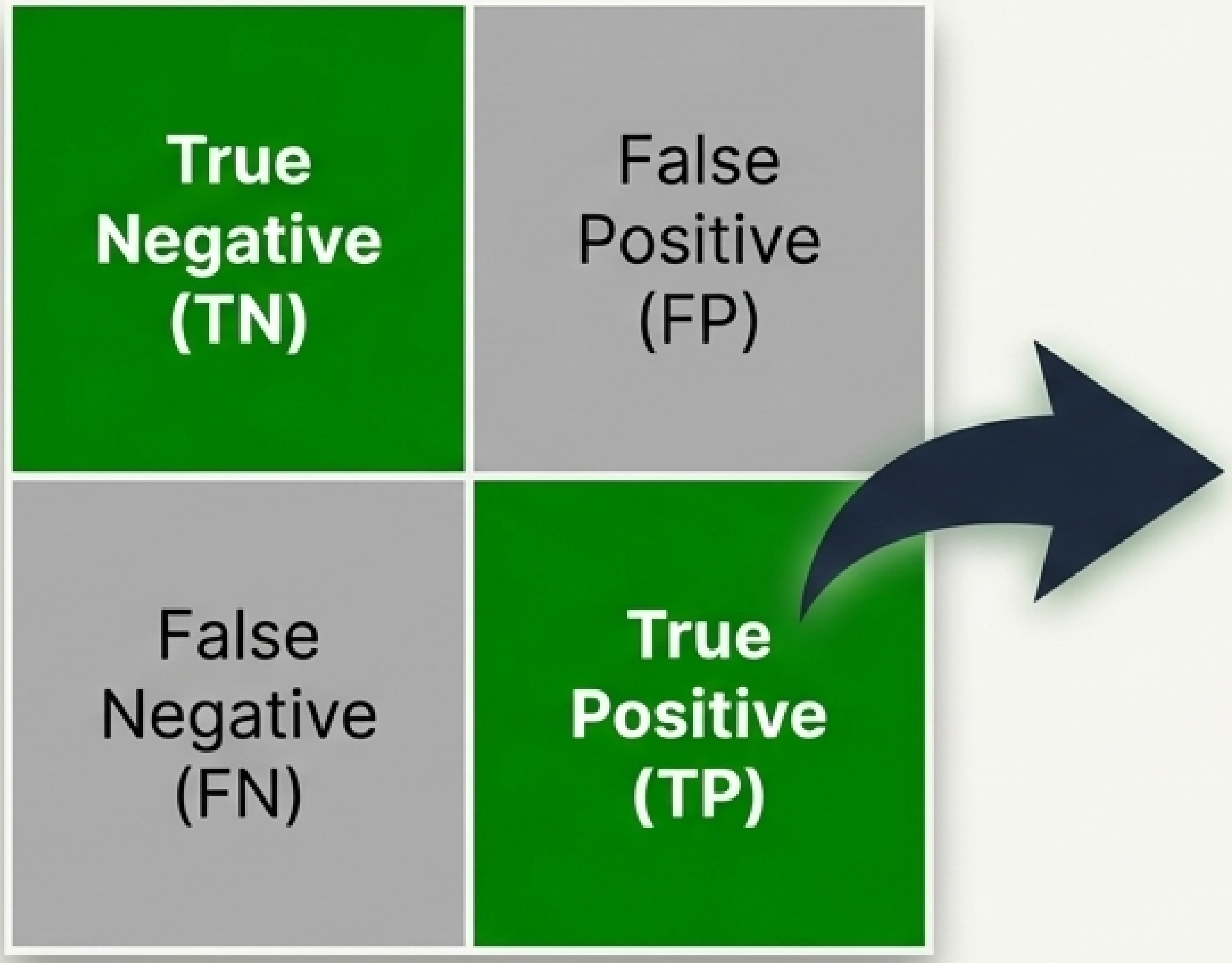
This tells you what the model predicted (1 = Positive, 0 = Negative).

Mapping the Four Quadrants

		Predicted Values	
		0	1
Actual Values	0	**True Negative (TN)** Predicted healthy, patient is healthy.	**False Positive (FP)** Predicted disease, patient is healthy.
	1	**False Negative (FN)** Predicted healthy, patient has disease.	**True Positive (TP)** Predicted disease, patient actually has it.

Calculating Accuracy directly from the Matrix

Accuracy is simply the sum of the correct diagonal divided by every number in the grid.



$$\frac{(\text{True Positives} + \text{True Negatives})}{\text{Total Predictions}}$$

The Taxonomy of Errors: Type 1 vs Type 2

Type 1 Error

False Positive

“The model cried wolf.”

It predicted a positive result (1), but reality was negative (0).

Type 2 Error

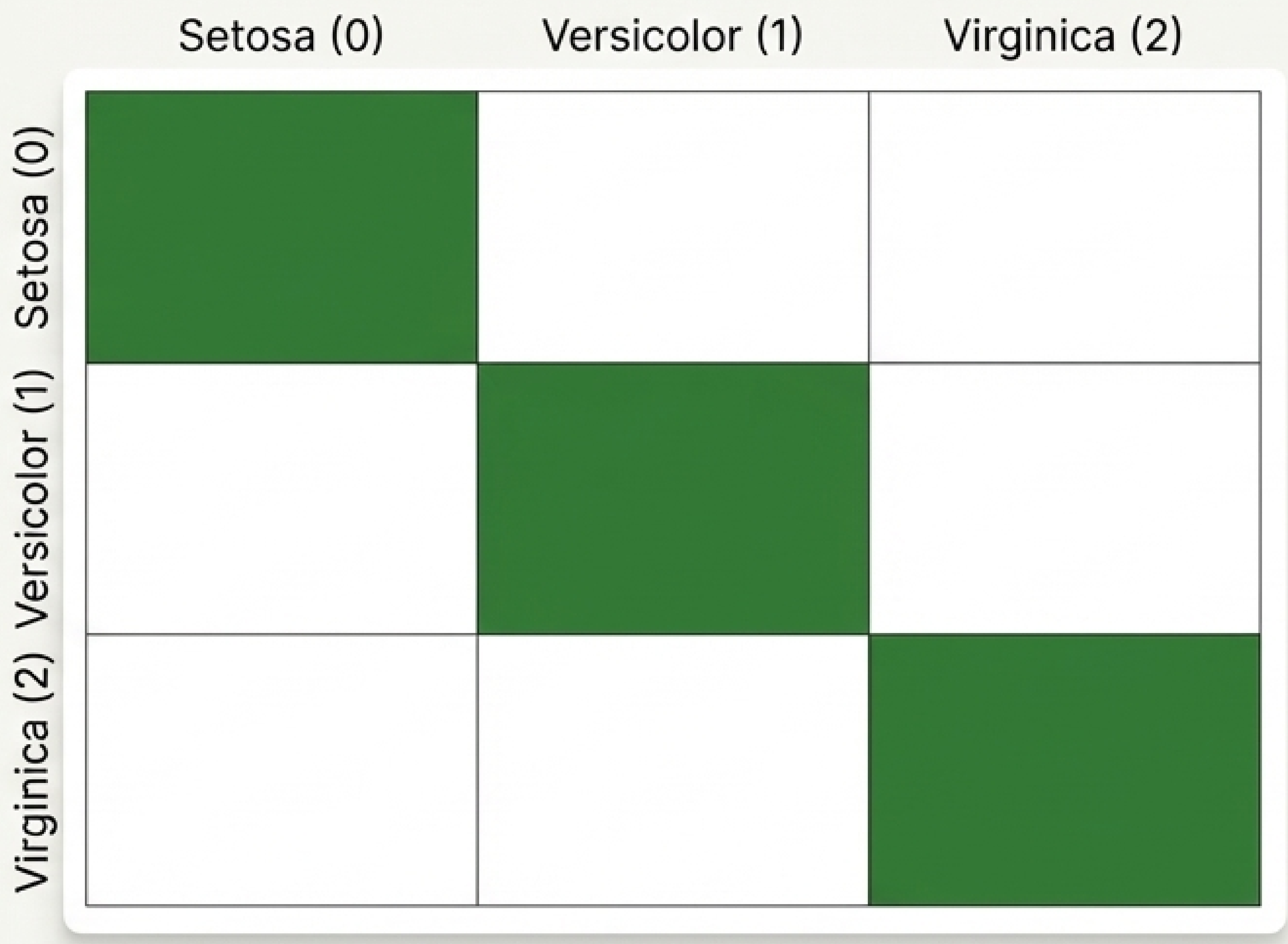
False Negative

“The silent threat.”

It predicted a negative result (0), but reality was positive (1).

Scaling Up: Multi-Class Matrices

The matrix adapts seamlessly to datasets with more than two classes. For three categories, the matrix simply expands to a 3×3 grid.



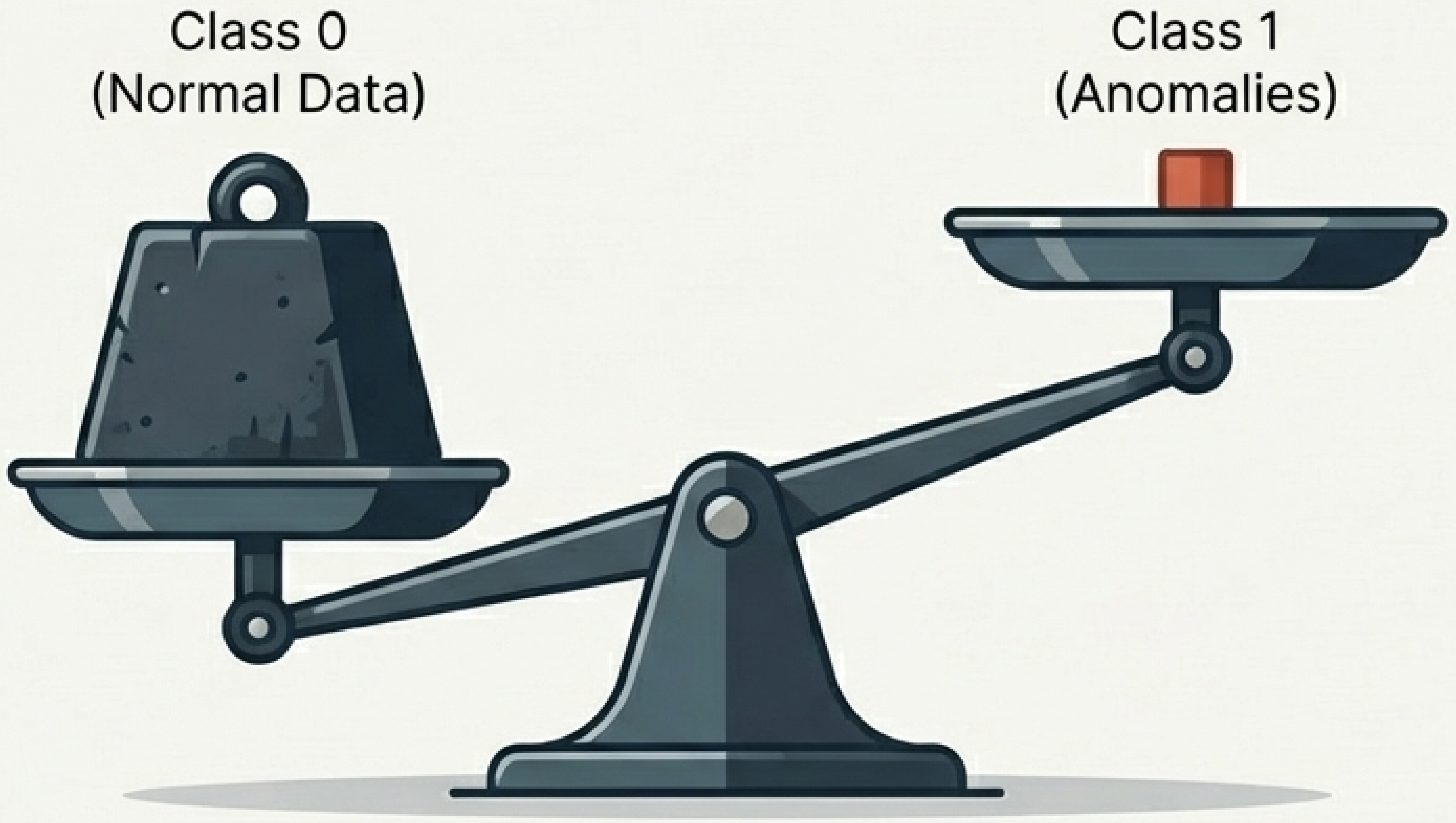
The Universal Diagonal Rule of Truth

Whether you have 2 classes or 10, the universal rule remains: True predictions always form a straight diagonal line. Every number outside that line represents an error.

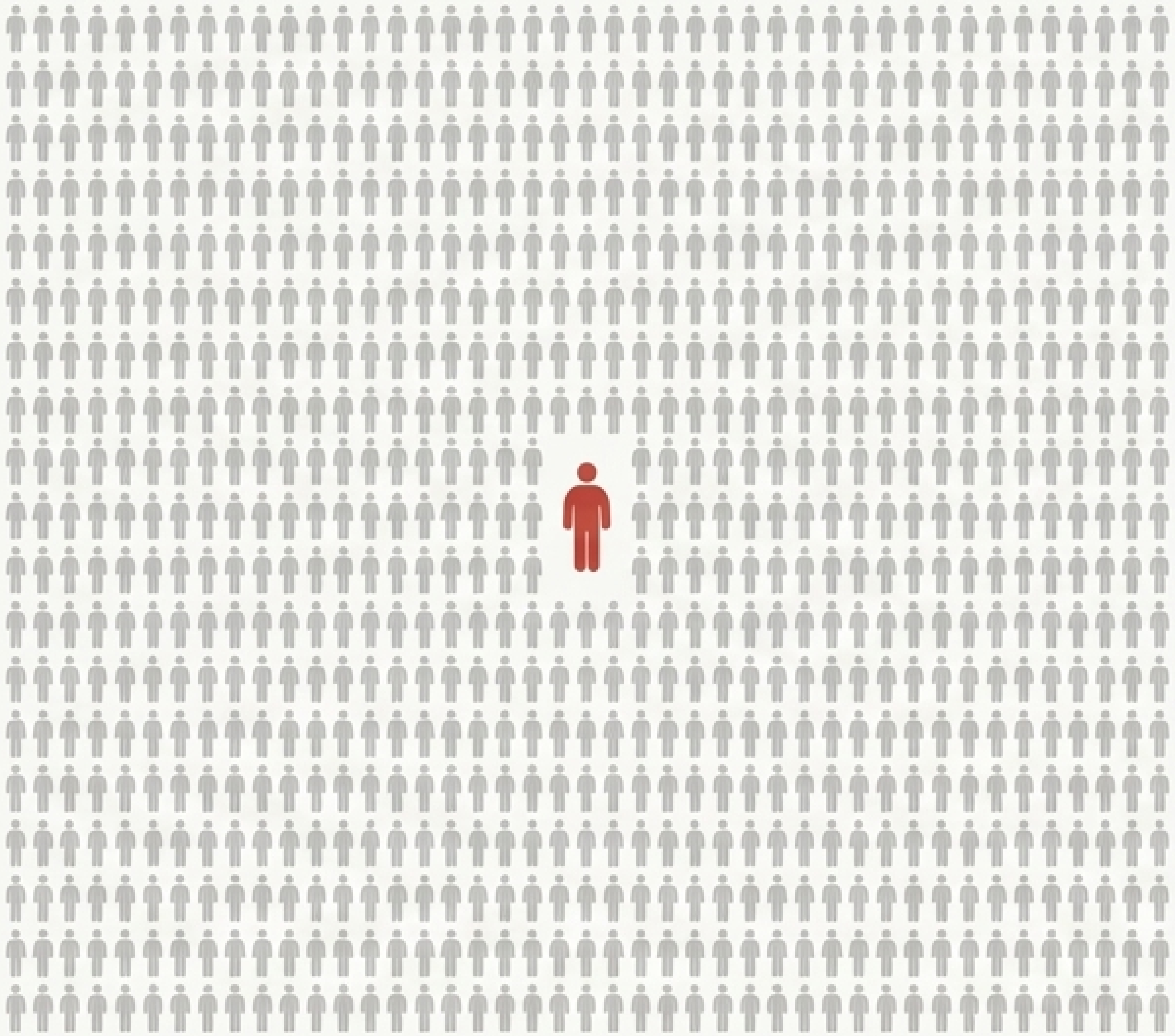
98		5		1				8	
5	85		2				1		
	2	92			1		2		
			93		9	3		1	3
	8	2		94		5			
1			3		96			3	1
			4	3		97	8		5
1	2	1			4		94		
2					3	1		92	4
		1	4			3	3		90

The Imbalanced Dataset Trap

Accuracy assumes a relatively equal 50/50 split in your data. But what happens when the reality of the data is overwhelmingly skewed in one direction?



The 99.99% Illusion



~~99.99%~~

Imagine an airport security model predicting terrorists. There are 99,999 normal passengers and 1 terrorist.

A lazy model that simply predicts “Normal Passenger” every single time will achieve 99.99% Accuracy—yet it completely fails its primary objective.

Where Accuracy fails, new tools emerge

Accuracy is the foundation. The Confusion Matrix is the ultimate diagnostic tool.
But when faced with highly imbalanced datasets, we must evolve our metrics again.